AMD Questions

Faced by : Ravi Bonthu

1) Tell me about yourself?

2)How you will verify an IP ?

3)In your Design suppose if an IP is added then **how can you verify that?**

D 4)Suppose you got a bug details from the verification team, then how will you fix it?

5) Tell me about your projects?

..... queries related to the projects

PD 6) tell me the synthesis flow and explain the Process?

PD 7) What are the violations you will face generally and how will you avoid those expalin?

8)suppose one black box is there, how will you verify it? means how will you write test plan?

Faced by : Dilip Prajapati

- **Questions related to my previous projects**
- **Details about verification environment and its individual components**
- **D: Questions related to Verilog for RTL coding and Behavioral coding**
- **Various ways to implement delays in Verilog**
- Various Verilog Constructs used in development of verification environment in verilog
- **Questions related to Behavioral Model development in Verilog**
- Asked in details SD Card's Behavioral Model which I developed in one of my project
- Questions related to debugging for bug full conditions

2nd Round

- **Questions related to my previous projects**
- **Details about verification environment of Ethernet MAC IP and its individual components**
- **Questions related to Ethernet Protocol**
- **Questions on Verilog for design and verification**
- Questions on validation/prototype and synthesis
- **Questions related to Arbitration logic implementation and verification**
- **General questions related to design and verification to measure individual's verification skill**

Faced by : Keyur Sanghvi

Round : 2

Interviewer : Narasimha

1. **What was asked to you in the first round?**

2. What is inheritance?

3. What are the different features of system verilog are very useful?

4. What is DPI?

5. Do you know the importance of import function

6. What is coverage? and functional coverage?

7. Tell me the difference between task and function?

8. Tell me the output of following code:

(1) module pra;

```
task ab(input int a);
   $display("%d t=",a,$time);
   #5;
   a = a+5;
   $display("%d t=",a,$time);
   #5;
   a= a+5;
   $display("%d t=",a,$time);
endtask

initial begin
   ab(5);
end
endmodule
```

(2) If i do

```
initial begin
```

```
ab(5);
#5;
ab(100);
end
```

9. When we call two different tasks, then does it create two different thread of it?

Ans:

10. What is inertial delay and transport delay?

11. What is blocking and non-blocking assignment

12. Two puzzle :

* **There is one function which can return any random no. Now I have one class packet in which there is one field call packet_length which can have range of 50-100 only. From a no which a function returns , how can i generate it in the range of packet_length.**

* **There is one box in which there are 10 white socks and 10 black socks. If a man can have only one pair than he can leave happily. What is the worst iteration possible for the right pair.**

Round : 1

Interviewer : Chetan

1. Describe yourself.

2. Do you have an experience of coding

3. You are mainly from computer engg. than why ASIC verification engineer

4. Do you know static variable

5**. What is the difference between static and global variable**

C 6. Why malloc and calloc is preferred over static

7. Do you know virtual function?

8. What is polymorphism?

9. Do you know fork and join

10. How process communication done in system verilog.

Faced by : Ravi Gupta

Round 1 & 2 :

Interviewer :
Give a brief introduction of yourself.

What projects have you worked upon.

Asked questions on what was my role in SOC verification project.

How does a DMA work?

Why do you really need a DMA?

What is a Stack?

How does a stack work?

How PUSH and POP instructions work in stack?

Why stack pointer is decremented when a PUSH operation is performed?

What are the different kind of delays in Verilog/VHDL ? Explain with examples.

Have you worked on OVM?

What is the difference between functional and code coverage? Explain an example

What is a covergroup?

What are cover bins?

Different types of coverbins?

Why do you require a methodology to verify blocks?

What are the advantages of system-verilog?

What are static variables? How are they different from global constants?

What are volatile and static variables?

What is a program block? Why is it needed?

What is a clocking block? Why is it needed?

What is a analysis port, how is it different from normal port?

Name some of the basic tasks/functions in TLM?

You have a microprocessor in its very basic stage, you need to verify MOV R, R instruction in it, and there are only 8, 32-bit registers implemented in the microprocessor. Explain how will you verify the instruction?

Faced by : Viral Patel

Round : 1

1. Introduction about your self.

2. what is blocking assignment and non blocking assignment

3. what is value of C in below code

1) A<= B

C<= A

2) A = B

C = A

4. find out the logic circuit of following code

```
if (sel ect== 1)
  y=A;
else
  y = B;
```

5. difference between task and function

6. difference between $display and $monitor

7. what is coverage? if i have a verification code which has not

coverage code than what happen?

**** 8. How i will verify memory.**

9. why seed is require?

10. Let I have a function myrandom() which return 32-bit random value

than how i generate values

between 0 to 99 .

11. Puzzle

I have three box..

one box has label APPLE.

another box has ORANGE

and remaining box has APPLE + ORANGE.

each box has wrong labels.

now i can peek only one fruit from any one box.

depending on that fruit ,make correct all label

Round :2

1. Tell me about your self.

2. What is your role and your work in current project?

3. My Educational background related questions like why you have two

years gap between Diploma and Degree.

4. Which languages do you know ?

5. What is the local and global variable?

6. What is the static variable?

C: 7. What is the pointer?

8. What is linked list?

9. how memory allocation done in linked-list?

10. Write a C program to reverse given string?

11. What is inheritance and why is it required?

12. do you know 8085 or 8086 microprocessor? What is pipe-lining ?

Faced by : Narendra Patel

Round 1 :

Interviewer : **Raj-Shekhar**

1) Share your experience with projects you worked on.

2) What is your role in each project?

3) Asked about the project architecture.

4) Questions related to protocol/specifications in detail.

Round 2 :

Interviewer : Raghu.

1) Share your experience with projects you worked on.

2) What is your role in each project?

3) Asked about the project architecture.

4) Implement the digital logic circuit to calculate no of 1 whether odd or even.

5) You have one (protocol based) interface which connects memory. It takes data from interface and stores into memory. How will you verify this? How will the architecture look like?

6) Above scenario, how will you verify if you don't have reference model?

7) How comparison will you use in above scenario? How you will use/implement scoreboard ?

8) What is parse memory?

9) implement the digital logic circuit to multiple two 4 bit numbers.

10) How you will implement digital circuit to check the register which has non-zero value?

Scripting: 11) How much comfort you with scripting?

Scripting: 12) what is hash? Give me one example of hash? What are advantages of hash over conventional array?

Faced by: Vikas Billa

1.what you know about AMD?

2.What is cache and main memory?

3.what is function and task?

4.some questions regarding AMBA AHB Lite (my previous project).

5.what is class?

6.Do you know c/c++/verilog?

7.what do you write in the test case? Explain?

8.What is Functional coverage and Code Coverage?

Round 1 and Round 2 combine questions

- **Give a brief introduction of youself.**
- **What was your roll in the last project?**
- **Have you made testplan?**
- **By whom the testplan was made?**
- **What is coverage?**
- **Types of coverage.**
- **Types of code coverage**

- **Have you write functional coverage?**
- **If code coverage is 100% and functional coverage is 80% what will you do?**
- **Do you know verilog?**
- **Do you know perl?**
- There is one file in which **Aashka** is written many times. If I want to replace it with Ravi what should I do?
- **Do you know C and C++ ?**
- **What is virtual class and what is virtual interface?**
- **What is modport?**
- **What is clocking block?**
- **What is program block?**
- **Difference between program block and clocking block.**
- **What is inheritance? Explain it with one example.**
- **What is polymorphism?**
- **Why randomization is needed?**
- **Why seed is used in randomization?**
- **What are the components of testcases?**
- What is gold report?
- **What is counter?**
- **You have a one counter in which there is a preloaded counter with up/down counter and output. Tell scenario through which I can verify the functionality of counter.**
- 2 puzzles were given. I was able to solve 1 only.
- You have joined the company in June,2011. Then why you want to come at AMD?

First interview was conducted by Vinita,

1.Rate yourself in SV,OOP & assembly.
2.Can you write assembly code.
3.Write assembly code to find no of 1's on odd and even positions of 32bit value.
4.What is cache memory.
4.Project related questions.
5.OOP advantages.
6.Explain pointer.
7.Explain polymorphism.
8.Overloading of functions.

Second interview was conducted by Vivek,

1. Why we use methodology.
2. Advantage of OVM.
3. Brief description about OVM components.
4. Addressing modes in micro processor.
5. Project related questions.
6. Assembly instructions.

7. How to verify instruction of processor.

Interview was taken by Chetan

- **Describe yourself.**
- **Do you have an experience of coding**
- You are mainly from computer engg. than why ASIC verification engineer
- **Do you know static variable**
- **What is the difference between static and global variable**
- Why malloc and calloc is preferred over static
- **Do you know virtual function?**
- **What is polymorphism?**
- **Do you know fork and join**
- **How process communication done in system verilog**

Aashka:

1. Tell me about LLM project.
2. What kind of assertions you have wrote in your project? give two or three examples.
3. Do you know Methodology?
4. Do you know OOPs?
5. What is virtual function?
6. What is inheritance?
7. What is cache and why it is used?
8. What is pipe lining?
9. If I have two class one is extended from other and both are having constructor then if I am instantiating the child class which constructor will be called?

Faced by Viren :

Interviewer : Sateesh

1. **Tell me about the projects you have worked on.**
2. **What is UVM and VMM?**
3. **Why to use methodology?**
4. **Explain your contribution in each of your projects.**
5. **Protocol related questions.**
6. What is structure in C?
7. What is union in C?
8. **Difference between Union and structure?**
9. Difference between pointer to an array and dynamic array?
10. **How you can create a dynamic array?**